Arista Access Node WFD-ACC-44 Whitefield POP Routing Protocol Flap Resolution SOP

Node: WFD-ACC-44 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6403

Description

This Standard Operating Procedure is specifically tailored to address the routing protocol flap failure on Arista Access Node WFD-ACC-44 located at the Whitefield POP. The observed failure domain is attributed to the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, manifesting as Alarm Code ALARM_6403. This failure domain occurs due to the inherent complexities of the routing protocol convergence mechanisms and the dynamic nature of the control plane on Arista Access nodes. In a typical scenario, BGP route flap damping, OSPF adjacency flapping, and ISIS LSP flooding can be triggered by various factors such as link failures, routing configuration errors, or even hardware malfunctions. The convergence of these factors on the Whitefield POP's Arista Access Node WFD-ACC-44 has resulted in the manifestation of the Alarm Code ALARM_6403, necessitating a swift and precise resolution.

Pre-requisite Checks

- Ensure the Whitefield POP is accessible and the Arista Access Node WFD-ACC-44 is reachable via SSH.

- Verify the redundant link status and confirm that the link in question is not a critical path.

- Check the current BGP, OSPF, and ISIS configurations to identify any potential misconfigurations.

- Verify the hardware temperature and power supply status of the Arista Access Node WFD-ACC-44.

Potential Root Causes

Cause	Probability
BGP route churn due to incorrect routing configuration	High
OSPF adjacency loss due to link failure or misconfiguration	Medium
ISIS LSP flooding due to hardware malfunction or high CPU utilization	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 4, main routing table version is 4, Description: This command verifies the current BGP configuration and displays the BGP table version, main routing table version, and the number of peers in established state.

Command: show ip ospf database, Expected Output: OSPF database has 10 entries., Description: This command displays the OSPF database and verifies the presence of OSPF routes.
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Command: show isis database, Expected Output: ISIS database has 5 entries., Description: This command displays the ISIS database and verifies the presence of ISIS routes.
```

Resolution Steps

Step 1: Step: Verify and correct BGP routing configuration, Command: show ip bgp summary | grep established, Description: Verify the number of peers in established state and correct any misconfigurations in the BGP routing configuration.

Step 2: Step: Verify and correct OSPF adjacency configuration, Command: show ip ospf neighbor, Description: Verify the OSPF adjacency status and correct any misconfigurations in the OSPF adjacency configuration.

Step 3: Step: Verify and clear ISIS LSP flooding, Command: clear isis database, Description: Verify the ISIS LSP flooding status and clear any ISIS LSP flooding issues.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 5, main routing table version is 5, Description: Verify that the BGP table version and main routing table version have incremented, indicating successful convergence.

Command: show ip ospf database, Expected Output: OSPF database has 10 entries., Description: Verify that the OSPF database has the expected number of entries, indicating successful convergence.

Command: show isis database, Expected Output: ISIS database has 5 entries., Description: Verify that the ISIS database has the expected number of entries, indicating successful convergence.
```

Roll-back Procedures

- Step: Restore BGP routing configuration to previous state, Command: copy run start, Description: Restore the BGP routing configuration to its previous state by copying the running configuration to the startup configuration.

- Step: Restore OSPF adjacency configuration to previous state, Command: copy run start, Description: Restore the OSPF adjacency configuration to its previous state by copying the running configuration to the startup configuration.

- Step: Clear ISIS LSP flooding, Command: clear isis database, Description: Clear any ISIS LSP flooding issues by clearing the ISIS database.